

Model

“A model in Angular is used to define the structure of application data. It helps in handling data between frontend and backend with type safety and better code organization.”

service

“A service in Angular is a TypeScript class used to perform common tasks like API calls, business logic, and data sharing, which can be reused across multiple components.”

“Instead of writing the same logic in multiple components, we place common functionality like API calls and data handling inside services.”

“This is a full-stack e-commerce web application developed using Angular for frontend and Spring Boot for backend.

In the frontend, Angular components are used to build different pages like login, products, cart, orders, and admin dashboard.

Services are used for reusable business logic and API communication with the backend using HttpClient.

Models are used to define the structure of application data such as Product, User, Cart, and Order.

Routing is used for page navigation, and SCSS is used for responsive UI styli

“Interceptor is used to automatically attach JWT tokens to HTTP requests for authentication and authorization.”

Guard

“Guard is used to protect routes and prevent unauthorized users from accessing certain pages.”

What is Angular?

Answer:

Angular is a TypeScript-based frontend framework developed by Google used to build dynamic and single-page web applications.

2. Why do you use Angular?

Answer:

Angular is used because it provides:

- Component-based architecture
- Reusable code
- Faster development
- Routing support
- Two-way data binding

- Better performance

What is SPA in Angular?

Answer:

SPA stands for Single Page Application.

In SPA, the page loads only once and updates content dynamically without refreshing the whole page.

Examples:

- Gmail
- Netflix

What is TypeScript?

Answer:

TypeScript is a superset of JavaScript that supports:

- Classes
- Interfaces
- Strong typing
- OOP concepts

Angular is built using TypeScript.

What is a Component?

Answer:

A component is the basic building block of Angular that controls a part of the UI.

A component contains:

- HTML template
- CSS styles
- TypeScript logic

What is Component-Based Architecture?

Answer:

Application is divided into reusable independent components.

What is decorator?

Decorators in Angular are used to provide metadata and define the behavior of classes such as components, services, and modules.”

“Decorators tell Angular how a class should behave.”

Simple Understanding

Angular ki ela telusthundi:

- idi component aa?
- service aa?
- module aa?

Because of decorators.

What is ngModule

“NgModule is used to group related components, services, directives, and pipes into one unit in Angular.”

What is a Module?

Answer:

A module is used to organize related components, services, directives, and pipes into one block.

Main module:

```
AppModule
```

What is Data Binding?

Answer:

Data binding is used to connect data between the component and the template.

Types of data binding:

- Interpolation
- Property Binding
- Event Binding
- Two-way Binding

What is Interpolation?

Answer:

Interpolation is used to display component data in HTML using double curly braces.

Example:

```
{{title}}
```

What is Property Binding?

Answer:

Property binding is used to send data from component to HTML element property.

Example:

```
<img [src]="imageUrl">
```

What is Event Binding?

Answer:

Event binding is used to handle user actions like click events.

Example:

```
<button (click)="submit()">
```

What is Two-Way Data Binding?

Answer:

Two-way data binding allows data transfer in both directions:

- Component to view
- View to component

Example:

```
<input [(ngModel)]="username">
```

What are Directives?

Answer:

Directives are used to modify the structure or behavior of DOM elements.

Types:

- Structural directives
- Attribute directives
- Custom directives

What are Structural Directives?

Answer:

Structural directives change DOM structure.

Examples:

- `*ngIf`
- `*ngFor`
- `*ngSwitch`

What are Attribute Directives?

Answer:

Attribute directives change appearance or behavior of elements.

Examples:

- `ngClass`
- `ngStyle`

What is `*ngIf`?

Answer:

`*ngIf` is a structural directive used for condition checking.

Example:

```
<p *ngIf="isLoggedIn">Welcome</p>
```

What is `*ngFor`?

Answer:

`*ngFor` is used to iterate over a collection.

Example:

```
<li *ngFor="let item of items">
```

What is a Service in Angular?

Answer:

A service is used to:

- Share data between components
- Handle API calls
- Store business logic

What is Dependency Injection?

Answer:

Dependency Injection is a mechanism where Angular automatically provides required service objects to components.

Advantages:

- Reusability

- Loose coupling
- Easier testing

What is Routing?

Answer:

Routing is used to navigate between different pages in an Angular application without reloading the browser.

What is Router Outlet?

Answer:

`<router-outlet>` is a directive used to display routed components.

What is Angular CLI?

Answer:

Angular CLI is a command-line tool used to create and manage Angular projects.

Commands:

```
ng new project-name
ng serve
ng generate component home
```

What are Lifecycle Hooks?

Answer:

Lifecycle hooks are methods that execute during different stages of a component lifecycle.

Examples:

- `ngOnInit`
- `ngOnChanges`
- `ngOnDestroy`

What is `ngOnInit`?

Answer:

`ngOnInit()` is called after component initialization.

Used for:

- API calls
- Initial setup

What is an Observable?

Answer:

An Observable is used to handle asynchronous operations like:

- HTTP requests
- User events
- Real-time data

What is a Pipe?

Answer:

Pipes are used to transform data in templates.

Examples:

- uppercase
- date
- currency

Example:

```
{{name | uppercase}}
```

Custom pipe

“Custom pipe is used to create our own data transformation in Angular.”

“First, we create a pipe using Angular CLI:

```
ng g p pipeName
```

“Then the pipe class implements `PipeTransform` interface and we write logic inside the `transform()` method.”

Example:

```
export class DemoPipe implements PipeTransform {  
    transform(value: string): string {  
        return value.toUpperCase();  
    }  
}
```

“Finally, we use the pipe in HTML using `|` symbol.”

Example:

```
{{name | demo}}
```

What is Lazy Loading?

Answer:

Lazy loading loads modules only when required.

Benefit:

- Faster application loading
- Better performance

What is Authentication?

Answer:

Authentication verifies whether the user is valid or not.

Process:

- User enters credentials
- Backend validates
- JWT token is generated
- Token is stored in localStorage/sessionStorage

What is Authorization?

Answer:

Authorization checks what permissions a user has after successful login.

Examples:

- Admin access
- User access

What are Guards?

Answer:

Guards are used to protect routes from unauthorized users.

Examples:

- CanActivate
- AuthGuard

What is an Interceptor?

Answer:

An interceptor is used to automatically modify HTTP requests and responses.

Uses:

- Add JWT token
- Handle errors
- Logging

What is RouterModule?

Answer:

Used to configure routes.

What is routerLink?

Answer:

Used for navigation in templates.

Example:

```
<a routerLink="/home">
```

FormControl

“FormControl is used to manage a single input field in Angular forms.”

FormGroup

“FormGroup is used to group multiple form controls into a single form.”

Template Driven Form

“Template-driven forms are managed mainly in the HTML template using ngModel.”

Reactive Form

“Reactive forms are managed in the component class using FormControl and FormGroup.”

Template Driven Form

“Template-driven forms are forms where most of the form logic and validation are handled in the HTML template using directives like ngModel.”

Reactive Form

“Reactive forms are forms where form controls and validations are managed in the TypeScript component class.”

```
form = new FormGroup({  
  name: new FormControl("")  
});
```

Template Driven

HTML-based

Simple forms

Reactive

TypeScript-based

Complex forms

Template Driven

Uses ngModel

Less scalable

Reactive

Uses FormControl/FormGroup

More scalable

What are Validators?

Answer:

Used for form validation.

Examples:

- required
- minlength
- maxlength

Which module is used for API calls?

Answer:

`HttpClientModule`

What is HttpClient?

Answer:

Used to make HTTP requests.

What is API Integration?

Answer:

Connecting frontend with backend services.

OBSERVABLES & RXJS

What is RxJS?

Answer:

Library used for reactive programming.

Common Pipes?

Answer:

- uppercase
 - lowercase
 - date
 - currency
 - percent
-

What is Change Detection?

Answer:

Angular automatically updates UI when data changes.

What is Virtual DOM?

Answer:

Lightweight copy of real DOM for performance optimization.

Why Angular over other frameworks?

Answer:

- Complete framework
- Strong architecture
- Better scalability
- Built-in routing and forms

Angular	React
Framework	Library
TypeScript	JavaScript
Built-in features	Third-party libraries

What is Signal in Angular?

Answer:

Signal is a reactive variable that automatically updates the UI when data changes.

Why do we use Signals?

Answer:

We use Signals for:

- Better performance
- Automatic updates
- Easier state management

How to create a Signal?

Answer:

```
count = signal(0);
```

How to access Signal value?

Answer:

```
count()
```

How to update Signal value?

Answer:

```
count.set(10);
```

OR

```
count.update(value => value + 1);
```

What is computed Signal?

Answer:

Computed signal creates value based on another signal.

Example:

```
double = computed(() => count() * 2);
```

What is effect()?

Answer:

`effect()` runs automatically when signal value changes.

Which Angular version introduced Signals?

Answer:

Angular 16.

Main advantage of Signals?

Answer:

Only affected UI updates, so performance improves.

Signals vs Observables

Answer:

Signals Observables

Simple Complex

Sync Async

State Data streams

Constructor	ngOnInit
Used for dependency injection	Used for initialization
Executes first	Executes after component creation

What is Local Storage?

Answer:

Local storage is used to store data in the browser permanently until manually cleared.

Used for:

- JWT tokens
- User data

What is Session Storage?

Answer:

Session storage stores data temporarily until the browser tab is closed.

Difference Between Local Storage and Session Storage

Local Storage	Session Storage
Permanent	Temporary
Available after browser restart	Cleared after tab close

What is JSON?

Answer:

JSON stands for JavaScript Object Notation.

It is used for data exchange between frontend and backend.

Example:

```
{
  "name": "Angular"
}
```

What is SCSS?

Answer:

SCSS (Sassy CSS) is an advanced version of CSS that provides additional features like:

- Variables
- Nesting
- Mixins
- Reusability

It helps write cleaner and maintainable styles.

What is Event Emitter?

Answer:

EventEmitter is used to send data from child component to parent component.

What is @Input?

Answer:

`@Input()` is used to send data from parent to child component.

What is @Output?

Answer:

`@Output()` is used to send data from child to parent component.

What is FormsModule?

Answer:

FormsModule is required for template-driven forms and ngModel.

What is subscribe() in Angular?

Answer:

`subscribe()` is used to receive observable data.

What is Standalone Component?

Answer:

Standalone components work without NgModule.

Introduced in newer Angular versions.

What is SSR in Angular?

Answer:

SSR (Server Side Rendering) renders pages on the server before sending to browser.

What is Bootstrapping in Angular?

Answer:

Bootstrapping means starting the Angular application.

What is AppModule?

Answer:

AppModule is the main/root module of Angular.

What is AppComponent?

Answer:

AppComponent is the first component loaded in Angular.

What is Selector in Angular?

Answer:

Selector is the component name used in HTML.

Example:

```
selector: 'app-home'
```

Usage:

```
<app-home></app-home>
```

What is Template in Angular?

Answer:

Template is the HTML part of a component.

What is styleUrls?

Answer:

styleUrls is used to connect CSS/SCSS files to a component.

What is templateUrl?

Answer:

templateUrl is used to connect an external HTML file to a component.

What is npm?

Answer:

npm stands for Node Package Manager.

Used to install Angular packages and dependencies.

What is package.json?

Answer:

package.json stores:

- dependencies
- scripts
- project details

What is Environment.ts?

Answer:

environment.ts stores configuration values like API URLs.

“This is a full-stack e-commerce application developed using Angular for frontend and Spring Boot for backend. Angular components are used to build pages like login, products, cart, orders, and admin dashboard. Services are used for reusable business logic and API communication using HttpClient. Routing is used for navigation, reactive forms are used for validations, interceptors are used for JWT token handling, and guards are used for route protection.”